

Cybersecurity Bayesian Network: Phishing Detection

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Contents

1	Task 1 - Build a Bayesian Network in pyAgram	1
1.1	Introduction	1
1.2	Methods	1
1.2.1	Variable Selection	1
1.2.2	Causal Structure	2
2	Task 2 - Learn a Bayesian Network in pyAgram	2

1 Task 1 - Build a Bayesian Network in pyAgram

1.1 Introduction

Phishing remains one of the most prevalent forms of cybercrime and continues to pose a significant threat to both individuals and organizations. Attackers typically disguise fraudulent emails as legitimate communications in an attempt to deceive users into revealing sensitive information, installing malware, or performing actions that compromise security. Despite advances in technical detection methods, phishing attacks remain highly successful because they exploit human behavior through social engineering techniques rather than purely technical vulnerabilities [AHNK21, JLN25].

Several studies have shown that phishing emails often contain identifiable indicators, such as suspicious sender information, misleading hyperlinks, poor grammar, and language designed to create urgency or invoke authority [PBP⁺16, Nai15, Ros25]. At the same time, user-related factors such as cybersecurity awareness strongly influence whether a phishing attempt ultimately succeeds [ZBDR⁺24].

Because phishing detection involves uncertainty and incomplete information, Bayesian Networks provide a suitable modelling framework. They allow probabilistic reasoning based on observable evidence while explicitly representing causal relationships between variables. Rather than making a deterministic decision, a Bayesian Network can estimate the likelihood that an email is malicious and how different pieces of evidence contribute to that assessment.

The objective of this work is to construct a Bayesian Network capable of estimating the probability that an email is a phishing attempt and the probability that it ultimately leads to account compromise. The model combines observable characteristics of an email with behavioral characteristics of the user, enabling both prediction and explanation of phishing-related risks.

1.2 Methods

1.2.1 Variable Selection

The variables included in the network were selected based on commonly reported phishing indicators in cybersecurity literature. Research has identified sender legitimacy, hyperlinks, grammatical quality, and urgency cues as important characteristics that users rely on when distinguishing phishing emails from legitimate communications [PBP⁺16, Nai15, Ros25].

The Bayesian Network consists of eight binary variables, shown in Table 1. All variables were modelled as binary variables in order to reduce model complexity and facilitate interpretation of the resulting probability distributions.

Variable	Description	Values
SenderReputation	Indicates whether the sender appears trustworthy or suspicious.	good / suspicious
EmailGrammar	Represents the quality of grammar and spelling used in the email.	good / poor
ContainsLinks	Indicates whether the email contains hyperlinks.	yes / no
UrgencyLanguage	Indicates whether the email contains urgent or threatening language.	yes / no
PhishingEmail	Indicates whether the email is a phishing attempt.	yes / no
UserAwareness	Represents the cybersecurity awareness level of the user	high / low
ClickLink	Indicates whether the user clicks a link contained in the email.	yes / no
AccountCompromised	Indicates whether the user's account becomes compromised.	yes / no

Table 1: Description and values of the selected variables in the Bayesian Network

1.2.2 Causal Structure

The network structure was manually designed according to causal assumptions derived from the phishing process.

SenderReputation, EmailGrammar, ContainsLinks, and UrgencyLanguage were modelled as direct causes of the variable PhishingEmail. This reflects the assumption that these observable characteristics provide evidence regarding whether an email is malicious. Previous studies have shown that phishing attacks frequently rely on suspicious sender identities, embedded links, and urgency-based persuasion strategies to increase user compliance [JLN25, Nai15, Ros25].

The variable ClickLink depends on both PhishingEmail and UserAwareness. This relationship captures the idea that phishing emails are more likely to trigger user interaction, while users with higher cybersecurity awareness are generally better able to recognize and avoid phishing attempts [ZBDR⁺24].

Finally, AccountCompromised was modelled as a direct consequence of ClickLink. This reflects a simplified attack chain in which account compromise becomes substantially more likely after interaction with a malicious link.

The resulting Bayesian Network models the progression from email characteristics to phishing detection, user behavior, and ultimately account compromise.

2 Task 2 - Learn a Bayesian Network in pyAgrum

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